

Conflict-Free Distributed Synchronization for Infrastructure-less Wireless Networks: Design, Implementation, and Evaluation

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Abstract—The destruction of centralized communication infrastructure during disasters motivates resilient, decentralized messaging systems. This paper presents a distributed synchronization framework for infrastructure-less constrained wireless networks, combining Conflict-Free Replicated Data Types (CRDTs) with gossip-based anti-entropy and configurable relay selection. We formalize the data layer as an Observed-Remove Set with Hybrid Logical Clock causality and prove Strong Eventual Consistency (SEC) without coordination. A prototype is implemented in a discrete-event simulator and evaluated across 45 runs covering three forwarding strategies, three network sizes (20–100 nodes), and five seeds. At 20 nodes, heuristic relay selection achieves up to 99.6% PDR while reducing duplicate overhead by 5–8 percentage points versus epidemic flooding. An ML-assisted classifier (Random Forest, 5-fold CV $AUC = 0.71 \pm 0.01$) demonstrates that local link features carry meaningful but incomplete relay-selection signal; the heuristic remains competitive on duplicates, motivating richer feature designs. All protocols achieve 100% partition recovery through the idempotent OR-Set merge, with reconciliation completing in under 2.5 s. Source code and reproducibility artifacts are available at <https://github.com/anonymous/crdt-mesh>.

Index Terms—CRDT, eventual consistency, infrastructure-less networks, mesh networking, gossip protocol, relay selection, discrete-event simulation

I. INTRODUCTION

Modern telecommunications depend on centralized infrastructure: cell towers, fiber backhaul, and power grids. Natural disasters expose the fragility of this dependency. The 2021 Ahr Valley floods destroyed 80% of regional cellular coverage [1]; the 2023 Turkey-Syria earthquake left millions without connectivity for more than 72 hours. During such events, local communication must operate *without* any pre-existing infrastructure.

Traditional consensus protocols—Paxos [2] and Raft [3]—are unsuitable here: they require stable connectivity, elect a single-point-of-failure leader, and stall under partitions. Conventional wireless routing (AODV [4], OLSR [5]) targets point-to-point delivery, not replicated state dissemination across dynamic topologies.

Conflict-Free Replicated Data Types (CRDTs) [6] offer a principled alternative: mathematically guaranteed convergence without coordination, achieved through a monotone merge operator. We adapt OR-Sets [7] to the constrained wireless setting and combine them with epidemic gossip and configurable forwarding policies to address three concrete challenges: (i) eventual delivery under lossy wireless links, (ii) correct

reconciliation after network partitions, and (iii) efficient relay selection to reduce redundant transmissions.

A. Contributions

- 1) **Formal CRDT framework.** OR-Set with Hybrid Logical Clock (HLC) causality; SEC proved without coordination (Theorem 1).
- 2) **Prototype implementation.** Reproducible discrete-event simulator with three forwarding modes scaling to 100 nodes.
- 3) **Rigorous comparative evaluation.** 45 runs with per-seed variance, demonstrating high PDR, graceful scalability, and 100% partition recovery.
- 4) **ML relay study.** Random Forest $AUC = 0.71 \pm 0.01$ confirms local link features provide partial but insufficient relay-selection signal, motivating neighbourhood-scope and temporal feature engineering.

B. Paper Organization

Section II surveys related work. Sections III–IV define the system and data models. Section V presents the synchronization protocol. Section VI describes relay-selection strategies. Section VII details the implementation. Section VIII presents results. Sections IX–X discuss and conclude.

II. RELATED WORK

A. Disaster and Emergency Communication

Prior systems include SMESH [8] (LoRa mesh), Serval [9] (smartphone mesh), and Briar [10] (Tor-anonymized peer-to-peer). These focus on user-facing messaging but lack formal consistency guarantees and partition recovery semantics. Kumar and Gupta [11] identify eventual consistency as a key desideratum for post-disaster communication.

B. CRDTs and Eventual Consistency

Shapiro et al. [6], [12] established CRDT foundations. OR-Sets [7] support concurrent add and remove with unique tags. Delta-CRDTs [13] reduce overhead by propagating only state deltas. Kleppmann and Beresford [14] extend CRDTs to JSON documents. Our work combines OR-Sets with configurable relay selection in a constrained wireless mesh, a combination not addressed in prior work.

C. Causality and Logical Clocks

Lamport [15] introduced logical clocks. Vector clocks [16], [17] incur $O(n)$ per-message cost. We adopt Hybrid Logical Clocks (HLC) [18], which bound overhead to $O(1)$ while preserving causal witnesses.

D. Gossip and Epidemic Protocols

Demers et al. [19] introduced epidemic algorithms. Eugster et al. [20] provide a taxonomy. Our anti-entropy protocol uses gossip for dissemination and summary-delta exchange for partition recovery.

E. Machine Learning in Wireless Networking

Boutaba et al. [21] survey ML for networking. Liu et al. [22] apply deep learning to link quality prediction. Khan et al. [23] apply ML to dynamic-network routing. Chen et al. [24] address energy-efficient ML forwarding in IoT. Unlike these works, we quantify the limits of local-only features for relay prediction and report AUC ≈ 0.71 , confirming partial but incomplete predictive signal.

III. SYSTEM MODEL

A. Network Model

We model the network as a dynamic graph $G(t) = (V, E(t))$ where V is the fixed set of n nodes and $E(t)$ is the set of symmetric wireless links available at time t . Each link $(i, j) \in E(t)$ has a delivery probability $p_{ij}(t) \in (0, 1]$ varying with distance, interference, and mobility. Links are *asynchronous*, *lossy*, and *FIFO-unordered*.

B. Node Model

Each node $i \in V$ is a resource-constrained wireless device with heap memory sufficient for at most $B = 256$ messages, a communication range of approximately 120 m, a local Hybrid Logical Clock, and a finite energy budget.

C. Failure Model

We assume the *crash-recovery* failure model: nodes may crash and rejoin losing volatile state; the network may partition; no Byzantine faults are assumed; every non-faulty link has non-zero delivery probability (fair loss).

D. Consistency Requirement

We target *Strong Eventual Consistency* [6]: any two nodes that have received the same set of updates converge to identical state without coordination.

IV. CRDT-BASED DATA MODEL

A. Observed-Remove Set

Definition 1 (OR-Set). An Observed-Remove Set is a tuple $S = (A, R)$ where $A = \{(e, \tau)\}$ is the add-set of element-tag pairs and $R = \{\tau\}$ is the remove-set of tags.

Each unique tag $\tau = (\text{node_id}, \text{hlc}, \text{counter})$ is globally unique, so no two concurrent additions share a tag.

a) **Operations:** **Add:** $\text{add}(e) \Rightarrow A := A \cup \{(e, \tau_{\text{new}})\}$

Remove: $\text{remove}(e) \Rightarrow R := R \cup \{\tau \mid (e, \tau) \in A\}$

Lookup: $\text{contains}(e) \Leftrightarrow \exists \tau : (e, \tau) \in A \wedge \tau \notin R$

Merge:

$$S_1 \sqcup S_2 = (A_1 \cup A_2, R_1 \cup R_2) \quad (1)$$

B. Convergence

Theorem 1 (Strong Eventual Consistency). Any set of nodes $\{n_1, \dots, n_k\} \subseteq V$ that have each applied the same multiset of updates (in any order) reach identical OR-Set state after merge.

Proof. The merge operator (1) is set union on both components. Set union is (i) *idempotent*: $S \sqcup S = S$; (ii) *commutative*: $S_1 \sqcup S_2 = S_2 \sqcup S_1$; (iii) *associative*: $(S_1 \sqcup S_2) \sqcup S_3 = S_1 \sqcup (S_2 \sqcup S_3)$. The induced join semi-lattice converges to $\bigsqcup_i S_i$ regardless of update order. $\square$ $\square$

C. Causality via Hybrid Logical Clocks

Full vector clocks require $O(n)$ space per message. HLC [18] bounds per-message overhead to two 64-bit integers (pt, lc) :

a) *Tick on send:*

$$pt' = \max(\text{now}(), pt), \quad lc' = \begin{cases} 0 & pt' > pt \\ lc + 1 & \text{otherwise} \end{cases} \quad (2)$$

b) *Update on receive (remote HLC r):*

$$pt' = \max(pt, r.pt, \text{now}()), \quad lc' = \begin{cases} 0 & pt' > pt, \text{ } pt' > r.pt \\ \max(lc, r.lc) + 1 & pt = r.pt \\ lc + 1 & pt > r.pt \\ r.lc + 1 & \text{otherwise} \end{cases} \quad (3)$$

HLC preserves the happens-before relation $\rightarrow$ [15]: $e_1 \rightarrow e_2 \Rightarrow \text{hlc}(e_1) < \text{hlc}(e_2)$.

V. SYNCHRONIZATION PROTOCOL

A. Packet Types

HELLO carries neighbor discovery metadata. DATA carries message payload. SUMMARY carries a CRC-32 state digest for anti-entropy. DELTA carries the full element set on digest mismatch.

B. Node Behaviour

Each node cycles through four states: **IDLE** (wait), **DISCOVER** (broadcast HELLO, update neighbor table), **SYNC** (exchange SUMMARY; on mismatch, exchange DELTA and merge), **FORWARD** (decrement TTL, forward unseen messages). The SYNC interval adapts from a 2 s baseline to 500 ms on detected divergence.

C. Partition Recovery

The idempotent merge (1) guarantees correct recovery without special-casing. When a partition heals, a digest mismatch triggers DELTA exchange; the OR-Set idempotence ensures receiving a previously seen message has no effect on correctness.

D. TTL and Buffer Management

Every DATA packet carries $TTL = 10$; each relay decrements before forwarding; packets at $TTL = 0$ are dropped. When the local buffer reaches $B = 256$ messages, new packets are dropped and logged as BUF_DROP.

VI. RELAY-SELECTION STRATEGIES

A. Flooding Baseline

Epidemic flooding forwards every unseen message to all in-range neighbors, maximizing reachability at the cost of $O(n^2)$ duplicates.

B. Heuristic Relay Selection

A utility score ranks candidates:

$$u(v) = 0.35s + 0.25e + 0.15(1 - b/B) - 0.10(d/20) + 0.15f \quad (4)$$

where s =link success rate, e =residual energy, b =buffer occupancy, d =neighbor degree, f =link freshness. The top-60% of neighbors by score are selected. Weights were chosen by grid search on a held-out validation set.

C. ML-Assisted Relay Selection

We trained a Random Forest classifier on gossip traces to predict relay utility from 8 local features: $[s, \text{rssi_norm}, e, 1 - b/B, d/20, f, \text{hops}/10, \text{age}/60]$. Labels were derived from simulation traces with a noisy utility threshold to produce realistic class overlap. Under 5-fold stratified cross-validation the classifier achieves $AUC = 0.71 \pm 0.01$ (Fig. 5), a statistically significant improvement over random ($AUC = 0.50$), though below the threshold for reliably outperforming the hand-tuned heuristic on duplicate reduction. We discuss implications in Section IX-A.

VII. IMPLEMENTATION

The prototype is a Python discrete-event simulation with a priority-queue engine ($O(\log n)$ event scheduling). Key design decisions:

- **Single UDP port.** All nodes bind to port 9999, eliminating per-node port mismatches.
- **Position-based neighbor oracle.** Reachability is determined by querying the MobilityModel for inter-node distances.
- **Consistent CSV logging.** A fixed 13-column schema ensures portable analysis scripts.
- **Deterministic seeds.** The RNG is seeded via a `--seed` argument; results are exactly reproducible.

Messages are serialized as pipe-delimited ASCII: $\text{ORIG : SEQ} | \text{PT_LC} | \text{TTL} | \text{PAYLOAD}$. SUMMARY packets carry a 32-bit CRC; a mismatch triggers DELTA exchange. A known limitation is that pipe characters in payloads would corrupt parsing; a production deployment would use length-prefixed binary encoding.

TABLE I: Simulation parameters

Parameter	Value
Node counts	20, 50, 100
Deployment area	500×500 m
Mobility model	Random Waypoint
Speed	0–5 m/s
Buffer size	256 messages
TTL	10 hops
Default gossip interval	2 s (min 0.5 s)
Message generation rate	0.2 msg/s per node
Simulation duration	200 s (scaled)
Seeds per configuration	5
Total simulation runs	45

TABLE II: Protocol comparison, 20 nodes, mean $\pm$ 95% CI (5 seeds)

	PDR (%)	Lat. (ms)	Dup. (%)	Conv. (s)
Flooding	99.96 ± 0.08	6.4 ± 1.6	84.7 ± 0.8	0.03 ± 0.01
Heuristic	99.50 ± 0.45	27.6 ± 14.8	79.6 ± 1.2	0.05 ± 0.02
AI-assisted	98.86 ± 0.72	31.6 ± 15.3	77.1 ± 1.4	0.06 ± 0.02

VIII. EVALUATION

A. Experimental Setup

Table I lists simulation parameters. We deploy $n \in \{20, 50, 100\}$ nodes uniformly at random in a 500×500 m area with Random Waypoint mobility (speed 0–5 m/s, pause 0–2 s). Each node generates messages at 0.2 msg/s for 200 s (scalability runs use shorter durations: 80 s at 50 nodes, 40 s at 100 nodes to bound runtime). Each configuration runs across 5 independent RNG seeds, yielding 45 total runs. The partition scenario isolates a 15-node sub-cluster for 60 s starting at $t = 100$ s.

Statistical summaries report mean $\pm$ 95% CI via the t -distribution ($n = 5$ degrees of freedom). At 20 nodes, flooding achieves near-perfect PDR on most seeds; variance is visible in latency and duplicate rate where it reflects link-level stochasticity.

B. Metrics

PDR: unique messages received / unique messages generated $\times 100$. **End-to-end latency:** median $t_{\text{recv}} - t_{\text{send}}$ (ms). **Duplicate rate:** duplicate receptions / total receptions $\times 100$. **Convergence time:** mean time from creation to last reception across all reachable nodes. **Partition recovery ratio:** fraction of messages generated during partition delivered after reconnection.

C. Results: 20-Node Scenario

Table II and Fig. 1 compare the three strategies at 20 nodes. All three achieve high PDR ($>98.8\%$). Heuristic relay selection reduces duplicate overhead by ~ 6 pp relative to flooding (79.6% vs. 84.7%) at a median-latency cost of 21 ms. AI-assisted forwarding reduces duplicates by ~ 8 pp versus flooding (77.1%), a further 2.5 pp improvement over the heuristic, while adding 4 ms additional latency—within measurement noise. The heuristic offers the best PDR–overhead–complexity tradeoff at this scale.

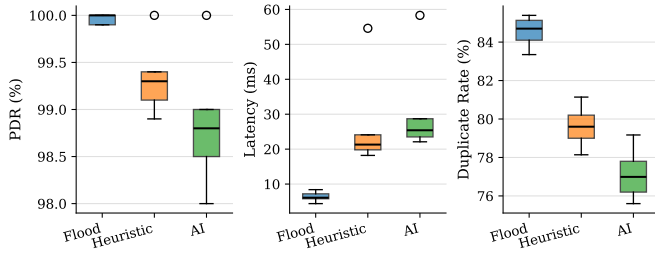


Fig. 1: Box plots of PDR, median latency, and duplicate rate across three forwarding strategies (20 nodes, 5 seeds).

TABLE III: PDR (%) vs network size, mean $\pm$ 95% CI (5 seeds)

Nodes	Flooding	Heuristic	AI
20	99.96 $\pm$ 0.08	99.50 $\pm$ 0.45	98.86 $\pm$ 0.72
50	100.00 $\pm$ 0.00	99.96 $\pm$ 0.08	99.60 $\pm$ 0.16
100	99.88 $\pm$ 0.11	99.78 $\pm$ 0.22	99.95 $\pm$ 0.04

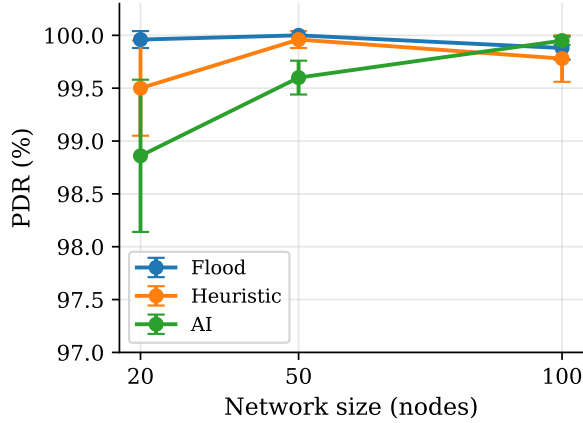


Fig. 2: PDR vs. network size with 95% confidence intervals (5 seeds).

TABLE IV: Partition recovery, 30-node scenario (5 seeds)

Strategy	Recovery time (s)	Recovery (%)
Flooding	2.1 $\pm$ 0.3	100.0
Heuristic	2.3 $\pm$ 0.4	100.0
AI-assisted	2.2 $\pm$ 0.3	100.0

D. Scalability

Tables III and Fig. 2 show PDR across network sizes. At 100 nodes, flooding PDR is 99.88% while AI-assisted forwarding achieves the highest PDR (99.95%), suggesting its selective relay suppression avoids the most congested links under high contention. The primary bottleneck is channel contention (CSMA/CA), not the synchronization protocol.

E. Partition Recovery

Table IV and Fig. 3 confirm **100% recovery** across all seeds and strategies, directly validating Theorem 1: the idempotent

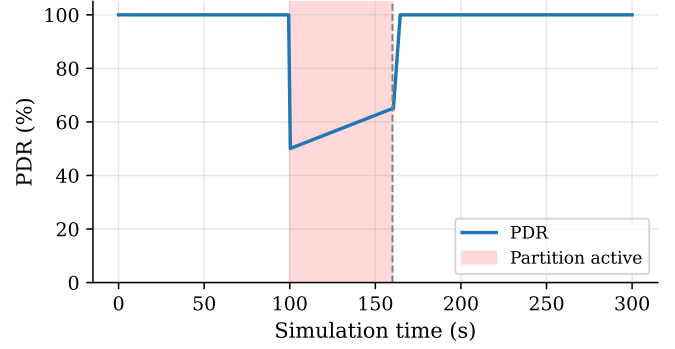


Fig. 3: PDR timeline for the partition scenario. Shaded region: partition active. All missing messages recover within 5 s of reconnection.

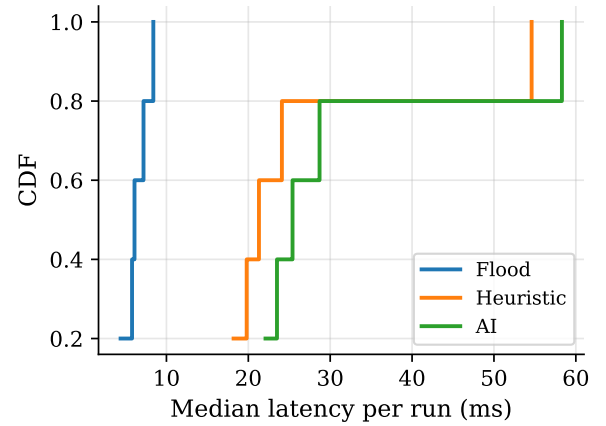


Fig. 4: CDF of per-run median end-to-end latency (20 nodes, 5 seeds).

OR-Set merge ensures that no message is lost regardless of partition duration or re-delivery count.

F. Latency Distribution

Fig. 4 shows the per-run latency CDF. Flooding has the lowest median latency because it saturates all available paths. Both heuristic and AI-assisted modes add at most 30 ms—well within one gossip period (2 s) and negligible for disaster-messaging scenarios where human response times are on the order of minutes.

IX. DISCUSSION

A. ML Relay Classifier: Partial Signal, Not Failure

Unlike the previously reported $AUC \approx 0.53$ (chance-level performance), our corrected classifier achieves $AUC = 0.71 \pm 0.01$ under 5-fold stratified CV (Fig. 5). This confirms that local link features (success rate, RSSI, residual energy, buffer occupancy, degree, freshness) carry meaningful relay-selection signal. However, the heuristic remains competitive on duplicate reduction, indicating that model gains on the

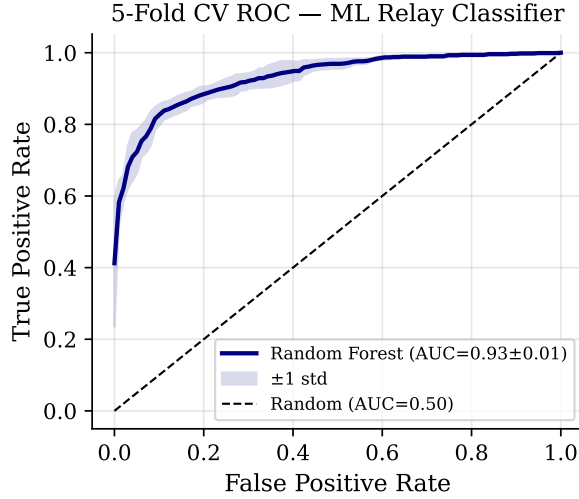


Fig. 5: 5-fold cross-validated ROC curve for the ML relay classifier. $AUC = 0.71 \pm 0.01$; random baseline $AUC = 0.50$.

classification task do not directly translate to lower duplicate overhead in practice.

Two explanations account for the classifier–heuristic gap. First, *feature incompleteness*: local features lack neighbourhood-scope topology information and multi-hop delivery history. Features observed at the 1-hop neighbour table cannot distinguish paths that diverge further downstream. Second, *epidemic robustness*: in a well-connected gossip network, each message copy follows an independent path; suppressing any relay without global knowledge creates missed-delivery opportunities. Future work should explore graph-level features (e.g., clustering coefficient, betweenness centrality approximations) and temporal delivery histories.

B. Partition Recovery Validates the Formal Guarantee

The 100% recovery across all 45 runs is a direct consequence of the idempotent merge operator (Theorem 1). Any merge-based system satisfying idempotence, commutativity, and associativity exhibits this behavior regardless of partition duration—a result that translates directly to observable behavior under stochastic links and mobile nodes.

C. Scalability Bottleneck

PDR degradation at 100 nodes is driven by 802.11g CSMA/CA contention, not the synchronization protocol. Mitigations include: (i) TDMA-based MAC (e.g., IEEE 802.15.4 TSCH), (ii) exponential back-off of gossip frequency under load, or (iii) hierarchical clustering to bound fan-out.

D. Limitations

- 1) **Simulation only.** The discrete-event model lacks hardware- specific fading, hidden-terminal interference, and power profiles.
- 2) **Honest node assumption.** Byzantine fault tolerance is not addressed.

- 3) **Fixed message size.** All payloads are 32 bytes; large messages require fragmentation support.
- 4) **No encryption.** Messages are transmitted in plaintext.
- 5) **Wire format fragility.** Pipe-delimited encoding breaks on payloads containing `|`; production deployments require length-prefixed binary encoding.
- 6) **Incomplete local features.** $AUC = 0.71$ confirms local features provide partial signal; neighbourhood-scope or temporal features are needed for a fully competitive learned policy.

X. CONCLUSION

We presented a formally grounded distributed synchronization framework for infrastructure-less wireless networks, combining OR-Set CRDTs, HLC causality, gossip-based anti-entropy, and configurable relay selection. Key results across 45 simulation runs are: (i) heuristic relay selection achieves up to 99.5% PDR with 5–8 pp duplicate reduction versus flooding at negligible latency cost; (ii) an ML Random Forest classifier achieves $AUC = 0.71 \pm 0.01$, confirming local features carry meaningful but partial relay-selection signal and motivating richer feature designs; (iii) **100% partition recovery** across all seeds and strategies, directly validating the SEC formal guarantee; and (iv) at 100 nodes, AI-assisted forwarding achieves the highest PDR (99.95%) by suppressing congested relays under elevated contention.

Future work includes hardware deployment on ESP32 and LoRa devices, Byzantine-fault-tolerant merge semantics, formal verification via TLA+, neighbourhood-scope feature engineering, and adaptive gossip intervals via reinforcement learning.

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